CHAPTER 4

MICROBES IN HUMAN WELFARE

CHAPTER ESSENCE AT A GLANCE:
Exploitation of microbial fermentation and enzymatic biocatalysis for industrial products, wastewater treatment, and biological pest control.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Household Microbes and Food Fermentation:** Lactobacillus and Lactic Acid Bacteria (LAB) convert milk to curd, improving nutritional quality by increasing Vitamin B12 and checking gut pathogens. Saccharomyces cerevisiae (Baker's yeast) ferments dough for bread, releasing CO2 to produce porous textures; traditional drink Toddy is produced by fermenting palm sap; Swiss cheese large eyes are produced by Propionibacterium shermanii releasing CO2; Roquefort cheese is ripened by Penicillium roqueforti.

• **Industrial Bioactive Molecules and Organic Acids:** Microbial organic acids: Citric acid (Aspergillus niger fungus), Acetic acid (Acetobacter aceti bacterium), Butyric acid (Clostridium butylicum bacterium), and Lactic acid (Lactobacillus). Industrial enzymes: Lipases in laundry detergents, Pectinases and Proteases for clarifying bottled fruit juices, and Streptokinase (from Streptococcus) as a clot-buster for patients suffering myocardial infarction.

• **Bioactive Immunosuppressants and Statins:** Cyclosporin A, an indispensable immunosuppressive drug in organ transplantation, is produced by the fungus Trichoderma polysporum. Statins, blood-cholesterol lowering agents that competitively inhibit HMG-CoA reductase involved in cholesterol synthesis, are produced by the yeast Monascus purpureus.

• **Sewage Treatment and Biocontrol Biofertilizers:** Wastewater secondary treatment utilizes aerobic microbial flocs (bacteria intertwined with fungal filaments) to consume dissolved organic matter, sharply reducing Biochemical Oxygen Demand (BOD); settled activated sludge undergoes anaerobic digestion generating biogas (methane, CO2, H2S). Biocontrol: Bacillus thuringiensis (Bt) controls caterpillars, Baculoviruses (Nucleopolyhedrovirus) provide narrow-spectrum insecticidal action; Biofertilizers: Rhizobium, Mycorrhizae (Glomus absorbing phosphorus), and Cyanobacteria (Nostoc, Anabaena).

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Bioactive Drugs} = \frac{\text{Cyclosporin A (Immunosuppressant: } \textit{Trichoderma})}{\text{Statin (Cholesterol-Lowering: } \textit{Monascus})}$$

Microbial Therapeutics in Modern Medicine

$$\text{Water Pollution Index} = \text{High Organic Waste} \rightarrow \text{High BOD} \rightarrow \text{Depleted Dissolved } O_2$$

Biochemical Oxygen Demand (BOD) Environmental Metric

$$\text{Clot Buster Action} = \text{Streptokinase (mathitStreptococcus)} \rightarrow \text{Lysis of intravascular thrombi}$$

Enzymatic Thrombolytic Intervention in Myocardial Infarction

$$\text{Biogas Production} = \text{Anaerobic Sludge} \xrightarrow{\textit{Methanobacterium}} CH_4\ (50\text{-}70\%) + CO_2 + H_2S$$

Methanogenic Bioconversion of Activated Sludge into Fuel

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** A high BOD (Biochemical Oxygen Demand) indicates high organic waste pollution in water; as wastewater treatment progresses and flocs digest organic matter, BOD decreases.

• **Alert:** Cyclosporin A is produced by a fungus (Trichoderma polysporum), while Statins are produced by a yeast (Monascus purpureus).

• **Alert:** Baculoviruses belonging to the genus Nucleopolyhedrovirus are narrow-spectrum, species-specific insecticidal viruses that have zero negative impacts on non-target organisms.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Name the microbial producing source and clinical application of: (a) Cyclosporin A, and (b) Statins. [Score: 2 Marks]

Answer & Key Valuation Steps:

• (a) Cyclosporin A: Produced by the fungus Trichoderma polysporum; used clinically as an immunosuppressive agent in organ transplant patients. **[1 Mark]**

• (b) Statins: Produced by the yeast Monascus purpureus; used as blood-cholesterol lowering agents by competitively inhibiting cholesterol synthesis enzymes. **[1 Mark]**

2. What is Biochemical Oxygen Demand (BOD)? How does the BOD value serve as an index of organic water pollution? [Score: 2 Marks]

Answer & Key Valuation Steps:

• BOD definition: The amount of oxygen that would be consumed if all the organic matter in one liter of water were oxidized by bacteria. **[1 Mark]**

• Index of pollution: A higher BOD value indicates higher organic pollution and poorer water quality; as clean water has lower organic matter, its BOD is very low. **[1 Mark]**

3. Explain the step-by-step secondary (biological) treatment of municipal sewage in wastewater treatment plants, detailing the role of microbial flocs. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Aeration tank: Primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it. **[1 Mark]**

• Floc formation and digestion: Aerobic microbes grow vigorously into flocs (masses of bacteria associated with fungal filaments to form mesh-like structures), which consume major organic matter, significantly reducing BOD. **[1 Mark]**

• Settling and digestion: Effluent is passed into a settling tank where flocs sediment as activated sludge; a small part is pumped back as inoculum, and remaining sludge is pumped into anaerobic sludge digesters producing biogas. **[1 Mark]**

4. Match the following commercial products with their producing microorganisms: (a) Citric acid, (b) Clot-buster Streptokinase, (c) Swiss cheese large eyes, (d) Butyric acid, (e) Bread dough leavening, (f) Acetic acid. [Score: 3 Marks]

Answer & Key Valuation Steps:

• (a) Citric acid: Aspergillus niger (fungus). **[0.5 Mark]**

• (b) Streptokinase: Streptococcus (bacterium). **[0.5 Mark]**

• (c) Swiss cheese: Propionibacterium shermanii (bacterium). **[0.5 Mark]**

• (d) Butyric acid: Clostridium butylicum (bacterium). **[0.5 Mark]**

• (e) Bread dough: Saccharomyces cerevisiae (Baker's yeast). **[0.5 Mark]**

• (f) Acetic acid: Acetobacter aceti (bacterium). **[0.5 Mark]**

5. Discuss the roles of Bacillus thuringiensis (Bt) and Baculoviruses (Nucleopolyhedrovirus) as biocontrol agents in integrated pest management. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Bacillus thuringiensis (Bt): Bacterial spores available in sachets are mixed with water and sprayed onto crops; insect larvae (e.g., caterpillars) eat the spores, endotoxin crystals dissolve in gut, killing larvae without harming beneficial insects. **[1.5 Marks]**

• Baculoviruses (Nucleopolyhedrovirus): Biological pathogens attacking insects and arthropods; they possess species-specific, narrow spectrum insecticidal applications with zero negative impacts on plants, mammals, birds, or non-target insects, making them ideal for IPM. **[1.5 Marks]**

6. Explain how Mycorrhizae (Glomus) and Cyanobacteria (Anabaena, Nostoc) function as biofertilizers to enhance soil fertility without chemical pollution. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Mycorrhizae (Glomus): Form symbiotic associations with plant roots, where the fungal hyphae absorb phosphorus from soil and pass it to the host plant, conferring resistance to root-borne pathogens and tolerance to salinity and drought. **[1.5 Marks]**

• Cyanobacteria: Autotrophic photosynthetic microbes widely distributed in aquatic and terrestrial environments; species like Anabaena, Nostoc, and Oscillatoria fix atmospheric nitrogen, serving as vital biofertilizers in paddy fields and adding organic matter to soil. **[1.5 Marks]**

7. A municipal wastewater sample has an initial BOD of 480 mg per liter. After prolonged aeration tank digestion, its BOD drops to 20 mg per liter. Explain the biological activity responsible for this drop and describe what happens to the settled activated sludge in anaerobic digesters. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Biological activity in aeration tank: Continuous mechanical aeration provides dissolved oxygen for vigorous aerobic growth of heterotrophic bacteria and fungi forming flocs. **[1.5 Marks]**

• BOD reduction: These microbial flocs mineralize and consume the dissolved organic pollutants as respiratory substrates, dropping the BOD from 480 mg per liter to 20 mg per liter. **[1 Mark]**

• Fate in anaerobic digesters: Settled activated sludge is pumped into anaerobic sludge digesters where anaerobic methanogenic bacteria (e.g., Methanobacterium) digest the aerobic bacteria and fungi. **[1 Mark]**

• Biogas production: During anaerobic digestion, microbes produce biogas (a mixture of methane, carbon dioxide, and hydrogen sulfide), which is harnessed as clean renewable energy. **[0.5 Mark]**